

ACTL2102 Tutorial

Week 3: Recurrence, Periodicity and Long-Run Behaviour

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Plan

- ▶ Main questions: **2.4 (parts 6–7), 2.13, 2.14, 2.15, 2.16**
- ▶ If time permits: 2.5, 2.6, 2.7, 2.17

Recurrent and Transient States

Let f_i be the probability that, starting in i , the process ever returns to i .

Recurrent: $f_i = 1$

Return is certain, so the process returns infinitely often.

Transient: $f_i < 1$

Each visit is the last with probability $1 - f_i$, so the number of periods spent in i is geometric,

$$(f_i)^{n-1}(1 - f_i),$$

with mean $\frac{1}{1 - f_i}$.

Counting expected visits gives an equivalent test: i is recurrent if $\sum_n P_{ii}^{(n)} = \infty$ and transient if that sum is finite.

That mean is what answers “expected number of periods spent in state i ”.

Identifying Them Without Computing f_i

You rarely need to compute f_i directly. Use the structure of the chain instead.

- ▶ An absorbing state is recurrent.
- ▶ A finite closed class consists of recurrent states.
- ▶ An open class, one that can reach states it cannot return from, consists of transient states.
- ▶ In a finite chain at least one state is recurrent; they cannot all be transient.
- ▶ All states of a finite irreducible chain are recurrent.

Recurrence is a class property

If $i \leftrightarrow j$ then either both are recurrent or both are transient, so classify once per class rather than once per state.

When f_i itself is wanted, decompose by first return:

$f_i = \sum_{n \geq 1} f_i^{(n)}$, where $f_i^{(n)}$ is the probability that the first return happens at step n .

Exercise 2.13: Return probabilities and expected time in state

Exercise 2.13: [dmc12, Solution] For the following transition probability matrix with states $i = 1, 2, 3$:

$$\begin{bmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1. Identify the transient and recurrent states;
2. Find the probability of returning to state i , given that the process starts in state i , for state $i = 1, 2, 3$;
3. What is the expected number of time periods the process will be in state i , given that the process starts in state i , for $i = 1, 2, 3$?

Period of a State

The **period** $d(i)$ is the greatest common divisor of all $n \geq 1$ with $P_{ii}^{(n)} > 0$. If $P_{ii}^{(n)} = 0$ for every $n \geq 1$, set $d(i) = 0$.

- ▶ $d(i) = 1$: the state is **aperiodic**.
- ▶ $d(i) > 1$: returns are only possible at multiples of $d(i)$.

The chain on $\{1, 2\}$ with

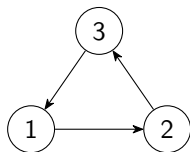
$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

A ± 1 random walk can only return to its starting point after an even number of steps, so again $d = 2$.

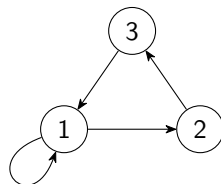
alternates deterministically, so $P_{11}^{(n)} > 0$ only for even n , giving $d = 2$.

A self-loop $P_{ii} > 0$ forces $d(i) = 1$, since 1 then belongs to the set. Period is also a class property.

Periodicity in Pictures



Returns to 1 take 3, 6, 9, ... steps, so $d = 3$.



One self-loop is enough: now 1 is a possible return time, and $\gcd(1, 3) = 1$, so every state is aperiodic.

Period is about **which** return times are possible, never about their probabilities. Changing the numbers on the arrows cannot change d ; adding or removing an arrow can.

Since period is a class property, that one self-loop makes states 2 and 3 aperiodic as well.

Positive Recurrence and Ergodicity

Recurrence says return is certain. It does not say return is quick.

- ▶ **Positive recurrent:** the expected return time m_{ii} is finite.
- ▶ **Null recurrent:** return is certain but the expected return time is infinite.
- ▶ **Ergodic:** positive recurrent and aperiodic.

Both positive and null recurrence are class properties.

The finite case

In a finite chain every recurrent state is positive recurrent. Null recurrence needs infinitely many states, so in most exercises “recurrent” and “positive recurrent” coincide and the only thing left to determine is the period.

Exercise 2.14: Period, positive recurrence, ergodicity

Exercise 2.14: [dmc13, Solution] For the following probability transition matrices:

$$\mathbf{P}_1 = \begin{bmatrix} 0 & 1/3 & 2/3 & 0 & 0 \\ 1/5 & 0 & 3/10 & 1/2 & 0 \\ 1/5 & 3/20 & 0 & 3/10 & 7/20 \\ 0 & 5/19 & 6/19 & 0 & 8/19 \\ 0 & 0 & 7/15 & 8/15 & 0 \end{bmatrix} \quad \mathbf{P}_2 = \begin{bmatrix} 0 & 1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 1/4 & 3/4 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1/8 & 7/8 & 0 & 0 & 0 \end{bmatrix}$$
$$\mathbf{P}_3 = \begin{bmatrix} 0.3 & 0.4 & 0 & 0.3 & 0 \\ 0 & 0.7 & 0.3 & 0 & 0 \\ 0 & 0.7 & 0.3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

1. What is the period of the states?
2. Which states are positive recurrent?
3. Which states are ergodic?

The Two Concepts

Limiting probability

$$\pi_j = \lim_{n \rightarrow \infty} P_{ij}^{(n)}$$

The chain forgets where it started. This requires the limit to exist.

For an **irreducible positive recurrent** chain, $\pi = \pi \mathbf{P}$ with $\sum_j \pi_j = 1$ has a unique non-negative solution, and

$$\pi_j = \text{long-run proportion of time in state } j = \frac{1}{m_{jj}}.$$

The distinction

If the chain is also **aperiodic**, these π_j are the limiting probabilities. If it is **periodic**, the stationary distribution still exists and still gives long-run proportions, but $\lim_n P_{ij}^{(n)}$ does not exist.

Stationary probability

$$\pi = \pi \mathbf{P}, \quad \sum_j \pi_j = 1$$

If $X_0 \sim \pi$ then $X_n \sim \pi$ for every n .

Solving $\pi = \pi \mathbf{P}$, and Using It

$\pi = \pi \mathbf{P}$ means $\pi_j = \sum_i \pi_i P_{ij}$: to find the weight on j , sum over everything flowing into j , so read down **column** j .

The equations are linearly dependent, so one is redundant. Drop any one of them and close the system with $\sum_j \pi_j = 1$.

- ▶ **Mean recurrence time**, $m_{jj} = 1/\pi_j$, the expected number of steps to return to j .
- ▶ **Long-run averages**: if a reward r_j is earned per visit to j , the average reward per step is $\sum_j \pi_j r_j$.

For average-reward questions, check what the reward attaches to. A payoff depending on the *transition* rather than the state needs $\sum_i \pi_i \sum_j P_{ij} r_{ij}$.

A Worked Stationary Distribution

A machine is working (state 1) or broken (state 2), checked daily:

$$\mathbf{P} = \begin{pmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{pmatrix}.$$

Read down column 1 for π_1 , and down column 2 for π_2 :

$$\pi_1 = 0.9\pi_1 + 0.4\pi_2, \quad \pi_2 = 0.1\pi_1 + 0.6\pi_2, \quad \pi_1 + \pi_2 = 1.$$

The first gives $0.1\pi_1 = 0.4\pi_2$, so $\pi_1 = 4\pi_2$. The second says the same thing, which is the redundancy. With the normalisation,

$$\pi = (0.8, 0.2).$$

Reading it back. The machine works 80% of days. Mean recurrence times are

$$m_{11} = 1/0.8 = 1.25 \text{ days}, \quad m_{22} = 1/0.2 = 5 \text{ days}.$$

The $1/\pi_j$ rule is just counting: a state visited on 20% of days comes round every 5 days on average.

Exercise 2.4: Long-run probabilities and expected visits (parts 6–7)

Exercise 2.4: [dmc3, Solution] For the following transition matrix with states $i = 1, \dots, 7$:

$$\begin{bmatrix} 0 & 0.8 & 0.2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0.6 & 0 & 0 & 0 & 0 & 0.4 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

1. Identify the absorbing states;
2. For states $i = 1, \dots, 7$, identify which states are accessible;
3. Identify the communicating classes;
4. Is the Markov chain irreducible?
5. Which states are recurrent and which are transient?
6. For each of the states 1, 3, 6, 7 would the long run probability of being in that state go to 0, be equal to a positive number less than one, or go to one, conditional on starting in that state.
7. For each of the states $i = 1, 3, 5, 7$ what would be the expected number of time periods that the process is in state i , given that the process starts in state i .

Exercise 2.15: Taxi driver: average profit per trip

Exercise 2.15: [dmc14, Solution] Ross: Chapter 4, Q52. A taxi driver provides service in two zones of a city. Fares picked up in zone A will have destination in zone A with probability 0.6 or in zone B with probability 0.4. Fares picked up in zone B will have destinations in zone A with probability 0.3 or in zone B with probability 0.7. The driver's expected profit for a trip entirely in zone A is 6; for a trip entirely in zone B is 8; and for a trip that involves both zones is 12. Find the taxi driver's average profit per trip.

Exercise 2.16: Health insurance scheme

Exercise 2.16: [dmc15, Solution] (IAA Subject 103 Exam April, 2000) The members of a health insurance scheme are classified as contributors or beneficiaries: a member who is a contributor in one period becomes a beneficiary in the next period if he or she becomes seriously ill and this happens with probability 0.1. The probability of a serious illness continuing into the next period is 0.2. The rules of the scheme specify that any member who is a beneficiary for three successive periods must become a contributor for the next period: if the illness still persists the member may thereafter revert to being a beneficiary.

1. Construct a discrete time Markov chain to model the health scheme.
2. Write down the transition matrix of the Markov chain.
3. Explain whether the Markov chain is irreducible, aperiodic or both.
4. Determine the stationary probability distribution of the chain.

Common Mistakes

Not this	This
$\pi_j = \lim_n P_{ij}^{(n)}$ always	periodic chain: π exists, the limit does not
$\pi_j = \sum_i P_{ji} \pi_i$ (along rows)	$\pi_j = \sum_i \pi_i P_{ij}$, down column j
$\pi_j > 0$ for a transient j	$\pi_j = 0$, however long the early visits last
one π for any chain	unique only with a single irreducible positive recurrent class
classifying state by state	per class: recurrence, period and positive recurrence are class properties

Thanks for coming!